NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR (Computer Science)

SPRING SEMESTER EXAMINATIONS 2025

Time : 3 Hours | Batch 2023 | Dated : 05-JUN-25

Max Marks : 60

Linear Algebra & Geometry - MT-272

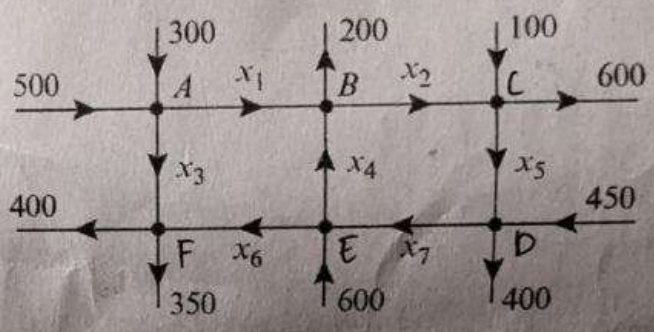
Instructions: 1): Attempt all questions.
2): All questions carry equal marks.

Q#1 [CLO 1] The accompanying figure shows a **network** of one-way streets with traffic flowing in the directions indicated. The flow rates along the streets are measured as the **average number of vehicles per hour**. **(12)**

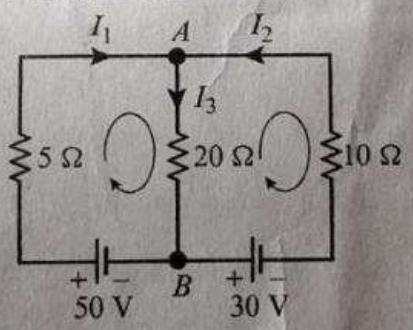
(a): Set up a **linear system** whose solution provides the unknown flow rates.

(b): **Solve the system** for the unknown flow rates.

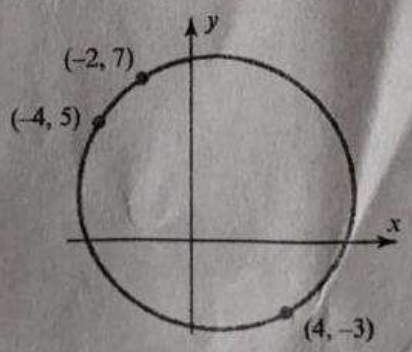
(c): Is it possible to close the road from A to B for construction and keep traffic flowing on the other streets? Explain.



Q#2 [CLO 1] (a): Determine the **currents I_1, I_2, and I_3** in the circuit shown in Figure. **(06)**



(b): Find the **coefficients a, b, c, and d** so that the circle shown in the accompanying figure is given by the equation $ax^2 + ay^2 + bx + cy + d = 0$. **(06)**



Q#3 [CLO 2] (a): Verify **Cayley Hamilton Theorem** for matrix A and then find A^{-1} and A^4. **(06)**

$$A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

(b): Write down any four **properties of eigenvalues** and prove them for matrix A. **(06)**

$$A = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ -1 & \frac{2}{3} & 0 \\ 5 & -8 & -\frac{1}{4} \end{bmatrix}$$

Q#4 [CLO 2] (a): Find the dominant **Eigen value** and the corresponding **Eigen vector** of matrix A by using the **Power method**. Find also the least Eigen value and hence the third Eigen value also. **(06)**

$$A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

(b): Express the **quadratic form** in the matrix notation $x^T Ax$, where A is symmetric, and discuss the **nature** of quadratic form. **(06)**

$$10x^2 + 2y^2 + 5z^2 - 4xy + 6yz - 10xz$$

Q#5 [CLO 1] (a): Derive the relation between **Direction Cosines.** **(06)**

(b): Find the **direction ratios, direction cosines** and **direction angles** of the vector $v = (3, 4, -6)$. **(06)**